

Virtual Fashion Mirror

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Abstract-Everyone loves to go shopping for clothes, except for the hassle that is the trial room experience. The Augmented Reality Fashion Display attempts to streamline the shopping process by removing all the troubles of trying on multiple clothes at a small cramped trial room. The product is capable of identifying a person, displaying clothes and accessories like watches on them in real time. This reduces the time spent waiting in queues and the cumbersomeness of trying on clothes that may have been worn by someone before you in the very same trial room. The applications for this technology can expand well beyond the scope of fashion stores and into the realm of fashion design itself enabling fashion designers to create models of their prototype designs and early concept art and test it out virtually before spending materials and resources to physically manifest these designs.

Keywords – *Augmented Reality, Occlusion, Vuforia, Image detection, Image Target.*

I INTRODUCTION

The Virtual Fashion Mirror is a new innovation of technology which bestows a new way to try on clothes as well as accessories without actually wearing them. Built with various features, it also provides information on the different sizes and colours of the garment that are available.. Apart from that, it gives suggestions on similar garments to promote in grown sales.

faster. Many shopping stores and malls are using this mirror as an interactive display in trial rooms. Customers, as well as retailers, are attracted towards the Virtual Fashion Mirror due to the beneficial outcomes it provides.

In one representation, a method of standardizing an image on a Smart Mirror display includes obtaining a moving body image. It includes generation of both right and left image overlays. The mirror displays the right and left image overlays

The rise of the modern fashion trends and the fashion industry have always been dependent on the growth of technology for production, design, consumption and distribution. Fashion is a merchandising and a marketing tool by nature. Merchandising is the essential soul and marketing is a perfect blend of fashion and an entrepreneurial intuition. In order to stay ahead on the fashion curve, the industry needs a mixture of compassion, art, marketing and business strategy. In the past 5 years, Augmented Reality and Artificial Intelligence have taken over the clothing industry by a storm where most designers and brands are heavily dependent on technology for production as well as sales processes. The digital age of Virtual and Augmented Reality in fashion has just started.

Our product enables users to try on a new attire in an environment created using an Augmented Reality Technology Stack. This product helps users to see how the apparels look in real-time without having to try them on manually. The Smart Mirror scans the height, shape and size of the customer in order to create the customer's digital twin. After the digital twin of the customer has been created, they can select different garments as well as various colours and patterns in an individual 3D catalog using hand gestures. If the customer is willing to buy an item, they can put it in the cart.

Using the concept of Augmented Reality, Smart Mirrors can make shopping easier and

on the captured image of the body. The method includes detection of the location of a left hand and the location of the right hand with their respective image overlay. Establishing at-least one point of reference on the captured image of the body corresponding to at-least one location of the image of the body has been included.

The Virtual Fashion Mirror applications don't just end within the consumer space but can also be used within the scope of

fashion design and technology spaces as well. Fashion designers who live in a very stressful world full of deadlines and galas lose valuable time by reworking physical prototypes, wasting raw materials and resources on early mock-ups of their dresses. Using the Virtual Fashion Mirror, fashion designers can directly translate the 3D models of their mock-ups and concept art for various designs into a Real Time Augmented Reality (AR) Environment where they can be viewed from all angles in a hassle-free way.

Customers are very particular about the clothes they buy, therefore; this allows customers to check where the outfit is loose or tight or where it sags down or juts out. In order to satisfy users, Augmented Reality is incorporated into the customer experience especially by a few forward thinking retail brands. Concentrating on the online platform as well as in stores, AR technology introduced this conclusion that works well even if the customer changes poses or moves around.

The lighting in the trial room is adjusted in order to enhance the user experience. This helps shoppers visualize how the outfits look at different timings and occasions. The user can have an idea about how the outfit feels and looks on him/her without actually trying on the dresses or standing in a queue.

II EXISTING WORK

A. *The impact of AR apps on purchase decision*

This paper describes that an online experiment was conducted with a large sample set of users, where the AR makeup retails app of a cosmetics company served as the stimulus. Using an approach in keeping with Javornik, two conditions existed within the experiment (2016a). When the option “augmentation” was chosen, participants interacted with the AR makeup, and when the option “non-augmentation” was chosen, the users interacted with the same app but with a lack of AR features. The effect of augmentation on purchase behaviour could be studied. An AR makeup app with the virtual reality mirror which was used by the experimental groups, allowed users to try on different makeup styles. The non-AR website allowed users to try on makeup products either through applying the makeup onto a model or an uploaded static image.

B. *Augmented Reality – Opportunity for Fashion by Alexandru TABU.CA*

This article visualises AR from the perspective of a software developer. The introduction of the paper aims to familiarize the reader with AR apps today, from code development, programming languages and choices. The result that the article presents is a new app developed for upgrading a previous AR solution. A step away from the 2D presentation, to a 3D presentation. The Burberry shop app uses an AR application in order to let users choose their skin colour, tone and colour of the nail paint..

C. *Top 5 Augmented Reality SDK's - Mark Baldwin*

In order to get started with Augmented Reality, a decision must first be taken on which Augmented Reality SDK to use. This is a very important preliminary step in the augmented reality application development process. Hence, a careful decision must be taken to ensure that a proper SDK is chosen for the application in development. As is the case with SDK's in general, the Augmented Reality SDK's are not created equally. Each of these SDK's have their own strengths and weaknesses. Proper research must be performed not just on the domain of the application but also the requirements of the application. Once the requirements of the application have been investigated in detail, the appropriate Augmented Reality SDK can be chosen.

Mark Baldwin has formed a ranking of the various Augmented Reality SDK's based on the search demand of each SDK along with the duration of time for which the SDK has been available for use in development. ARKit by Apple and ARCore by Google are the relatively new offerings compared to the other SDK's present on this list. Other SDK's such as Vuforia, Kudan, and Wikitude have been available to use for years. The particular AR SDK used for this application is Vuforia. The Vuforia platform is the most feature packed of all the AR SDK's on the offering, allowing for developers to achieve their visions without worrying about limitations imposed by technology. It supports features such as Image Target Based Augmented Reality which is supported by its superior, stable and technically efficient computer vision based image recognition capabilities. Vuforia also provides support for cross platform apps with support for iOS, Android, and Unity 3D. Now developers can write a single native app that can be used by the majority of users across the most popular segments of the handheld devices market.

I. DESIGN OF VFM

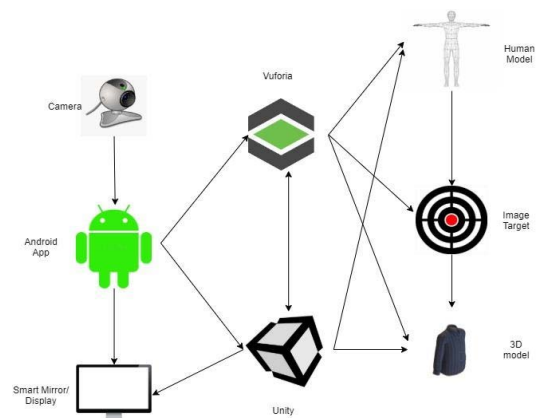


Figure 1.1 Design of VFM

The Virtual Fashion Mirror is powered by the Unity 3D Engine. While the Unity 3D engine is generally used for the development of 2D and 3D video games, when paired with Vuforia, it is also capable of performing powerful 3D augmentation.

Unity is capable of building cross-platform applications that can easily run on many platforms. However, Vuforia does not support all these platforms. Vuforia currently has support for Unity 3D, Android and iOS. Vuforia provides excellent Image Detection and Augmentation capabilities.

Vuforia provides a Marker Based Augmented Reality solution. By looking for and detecting a particular Image Target (a type of marker), Vuforia can then augment any 3D model to a position relative to the Image Target. While this may allow us to easily display a 3D model of some accessory or clothing item, in order to place them as accurately on the user, occlusion is performed.

In order to perform occlusion, a 3D human model is required. We used MakeHuman, an open source software that creates 3D human models with ease. This 3D model is used as the basis for occluding the 3D clothes and accessories on top of the user. The 3D human model is masked using a mask shader so that it is not visible to the user. Unity places this 3D model just behind the image target after which occlusion is done with the desired clothing item/accessory.

III IMPLEMENTATION

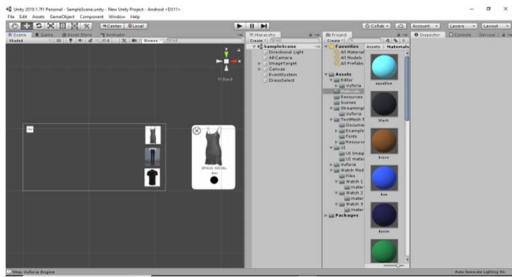


Figure 2.1 Unity 3D Development Environment

The application was developed with the use of Unity 3D Engine. The Unity 3D Engine is capable of deploying applications to multiple target platforms, making them cross platform compatible. The Vuforia extension is used along with Unity for the purpose of Image Target Recognition and augmenting the 3D models as well.

The implementation of the app utilizes the image recognition capabilities of the Vuforia extension of the Unity 3D Engine. Care is taken to prepare image targets that have several points of recognition to enable Vuforia to identify the image target with as little latency as possible.

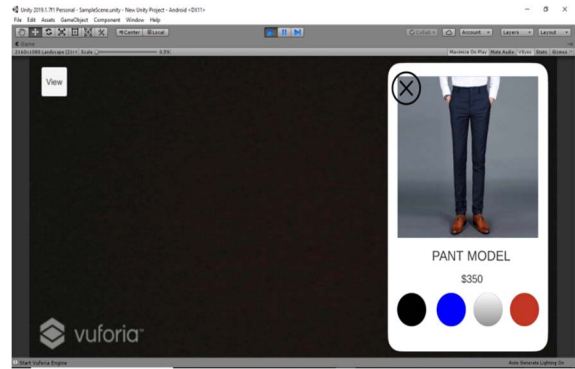


Figure 2.2 UI Developed in Unity 3D Engine

Once the image target is recognized, Vuforia augments the 3D model of the desired clothing item or accessory on top of the user. Vuforia uses a marker based system for identification of Image Targets and Augmentation.

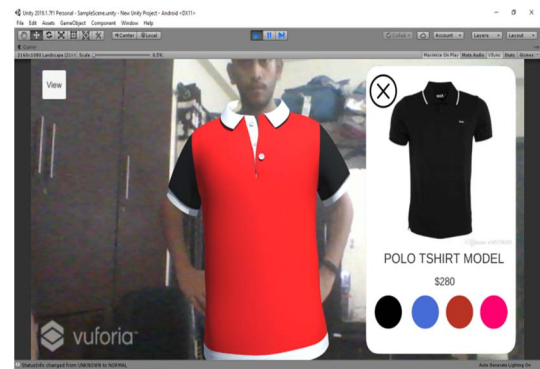


Figure 2.3 Polo T-shirt being augmented in real time

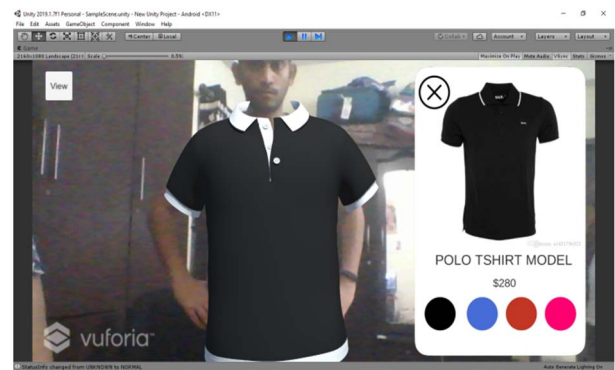


Figure 2.4 Changing the colour of the Polo T-shirt in real time

The 3D models were created in Blender, a 3D modelling tool that is prominently used. The 3D models were created with a high polygon count. MakeHum

an, a software designed to make 3D character (human) creation fast and simple, was used to create standard male and female models on which occlusion was performed with the 3D dress models created in Blender.

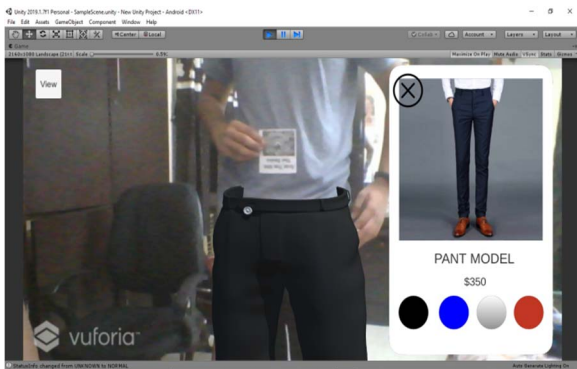


Figure 2.5 Pants being augmented in real time

The MakeHumanExchange format was used to communicate between the MakeHuman and Blender applications. This allowed for 3D rigging of the character models with the help of Blender's rigging and animation tools. The character models were created with a high polygon count and were provided with a basic skeleton rig. While MakeHuman provides many different standard poses in a MotionCaptureFormat, Blender was used instead to create custom poses by manually rigging the skeleton rig of the MakeHuman character models. After rigging, the models are then imported as blend files into a Unity Project.

IV FEATURES OF VFM

1. *Fully Localized Augmentation*

The Virtual Fashion Mirror is capable of performing augmentation completely locally on a device, foregoing the need of an active internet connection during augmentation. The reason this design choice was made was to avoid security and privacy concerns that arise as a consequence of having footage recorded and being sent to a server for further processing.

2. *Freedom of Movement*

Since the augmentation provided by Vuforia is a marker based, it allows for the user to move around the view of the camera and the clothes/accessories will be augmented on the user provided the user and the image target remain in view of the camera.

3. *Fast Detection*

Vuforia provides excellent Image Target detection, provided the Image is rated 5 stars by Vuforia, deeming it augmentable. Care is taken to ensure that all image targets are unique and possess several points of recognition to allow for speedy detection followed by augmentation.

4. *Real Time Changes*

Users can change the color of certain clothes/accessories while viewing them in augmented 3D, allowing them to see the changes happening in real time. This not only saves time but also serves as a point of attraction.

IV CONCLUSION

With the Virtual Fashion Mirror, retailers will shortly have a powerful new artillery in the battle to transform and renovate the in-store experience. In order to attract customers to use the Virtual Fashion Mirror, executives should place the mirrors at the center of the shop. The executives should ensure their technology infrastructures are up to mark so as to fulfil terms of supporting interactive and rich-media experiences. Retailers that prepare now will be in the best position to obtain the opportunity of growing revenues from consumers who spend huge amounts when they use more than one channel to shop.

The Virtual Mirror is made with the aim of providing customers the most fulfilling and doubt free shopping. The Virtual Mirror units can be easily merged with the store design. The Mobile application and Website can easily be accessed by the client and provides a hassle free shopping experience.

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